



Lecture 16:

VLSI Delay-Locked Loop Circuits

Contents



This topic: VLSI Delay-Locked Loop (DLL) Circuits

- Introduction
- VLSI DLL Architectures
- VLSI DLL Circuit Implementations

Introduction



- DLLs are special type of phase-locked loop (PLL) circuits
- DLLs are based on the voltage control delay line (VCDL) for delay compensation
- DLL only delays the input signal. It does not generate it.
 - more relaxed trade-offs among gain, bandwidth, and stability
 - less jitter than PLL using VCO
- In addition to phase alignment, DLL can also provide precise spaced timing edge even in the presence of PVT variations

Typical DLL Circuit Applications

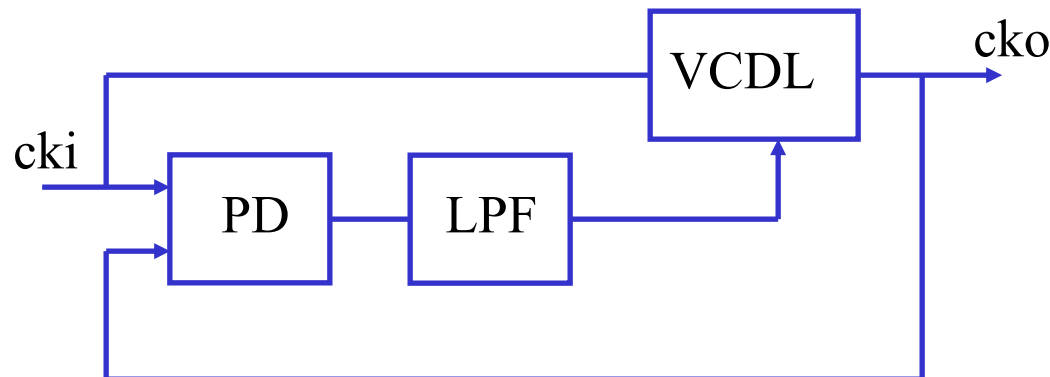


- I/O interface of to hide clock distribution delay
- Clock phase manipulation
- Controllable signal path delay
- Multi-phase clock generation
- Clock multiplication

Basic DLL Architecture



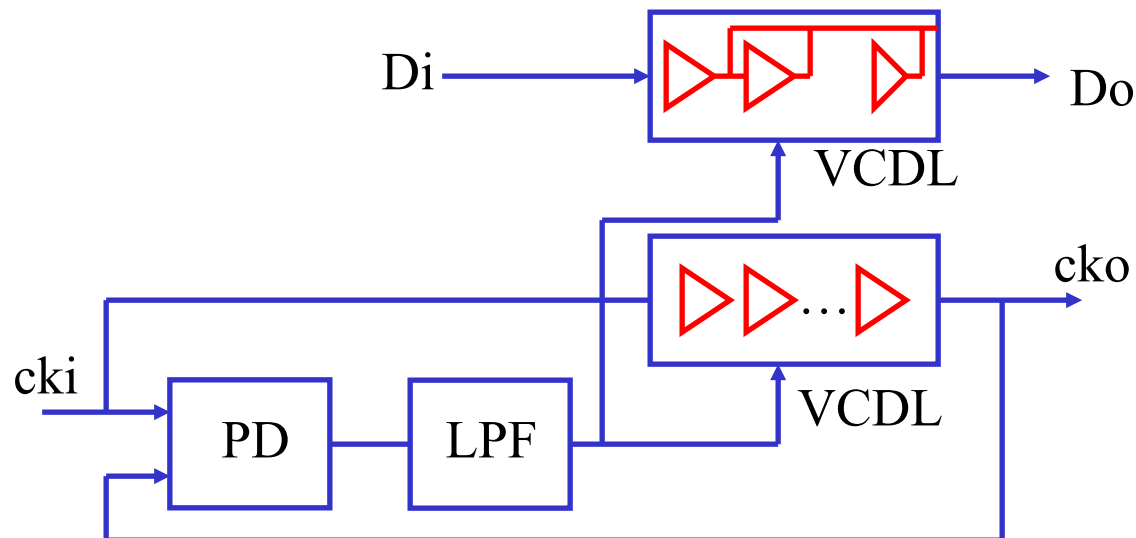
- A Basic DLL circuit includes the following building blocks
 - Voltage Control Delay Line (VCDL)
 - Phase Detector (PD)
 - Loop Filter (LPF)



Master Slaver (M-S) DLL Circuits

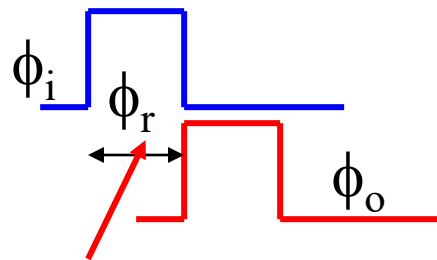
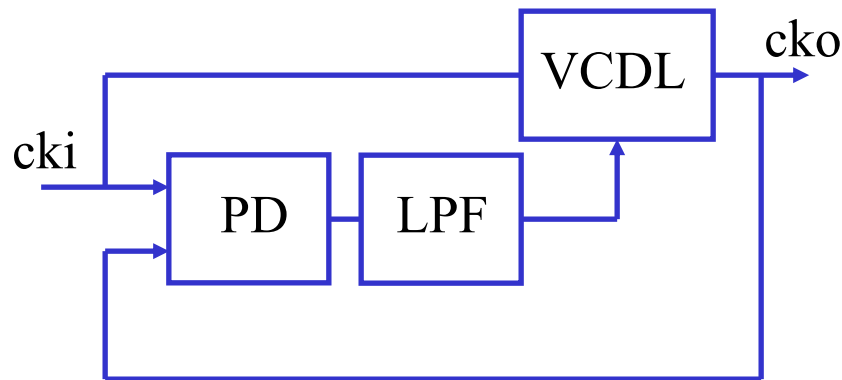


- A Master-Slaver DLL structure consists of
 - A Master in-loop delay line for delay reference
 - Replica slaver delay line(s) for signal or clock delay manipulation
 - Programmable offset may be introduced

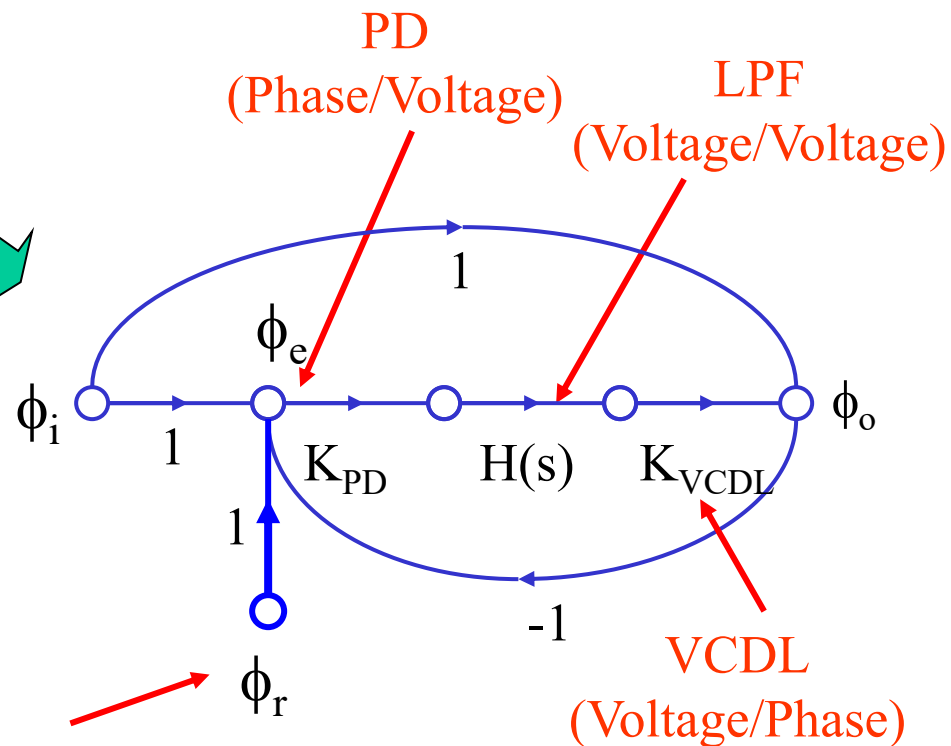


Linear Model of Basic DLL Circuit

- Small signal model (Linear model)
- Mixed Phase/Voltage (Current) domain system
- A Z-to-S transformation approximation



Delay Time reference

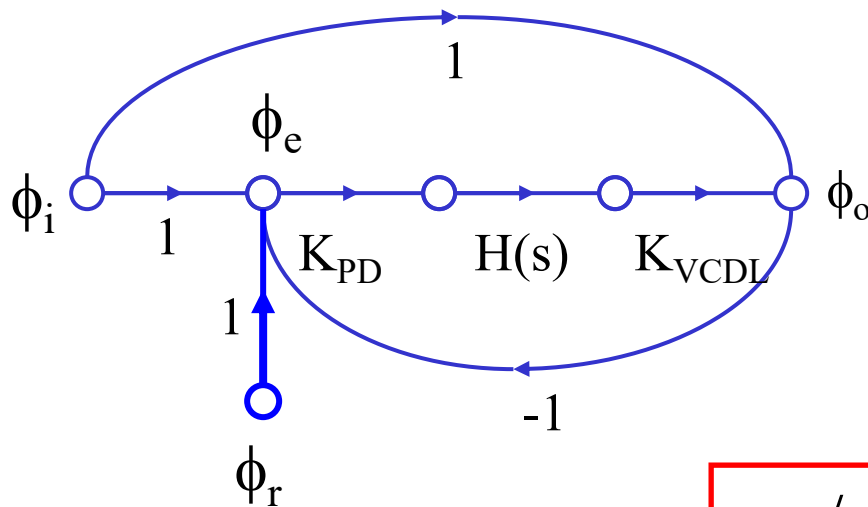


Delay Time reference

Linear DLL Transfer Function Models



- Delay and Phase Error Transfer Functions



Delay line length

$$\frac{\phi_o - \phi_i}{\phi_r} = \frac{K_{PD}K_{VCDL}H(s)}{1 + K_{PD}K_{VCDL}H(s)}$$

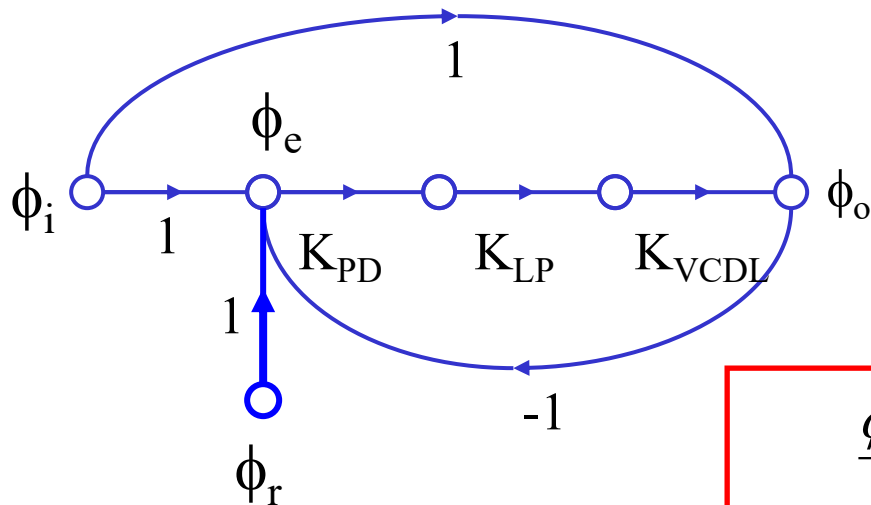
$$\frac{\phi_e}{\phi_r} \equiv \frac{\phi_r + \phi_i - \phi_o}{\phi_r} = \frac{1}{1 + K_{PD}K_{VCDL}H(s)}$$

Delay Tracking error

0th Order DLL Transfer Function



- $H(s) = K_{LP}$
- Phase error is inversely proportional to the loop gain



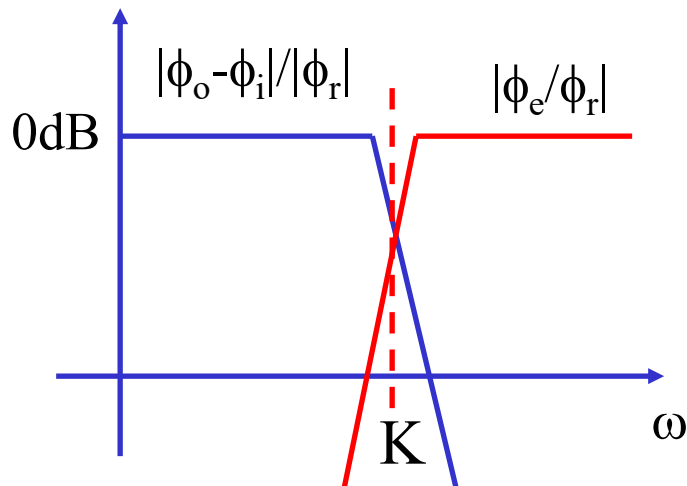
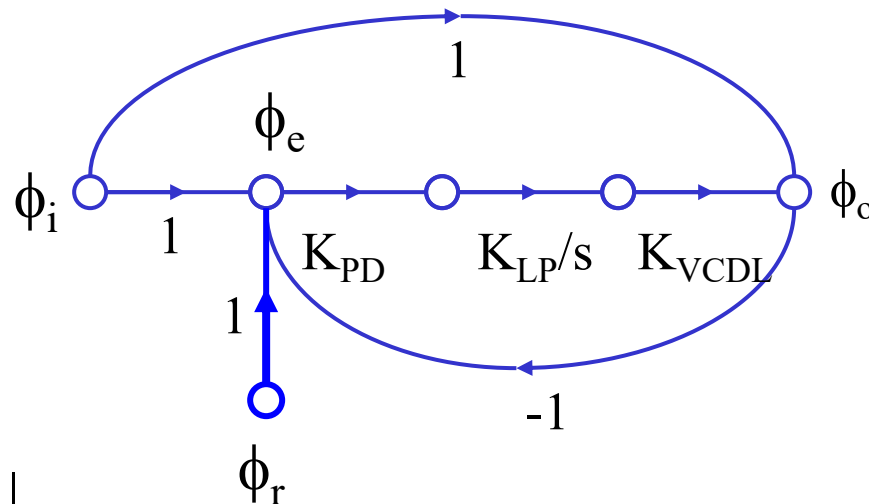
Loop Gain

$$\frac{\phi_o - \phi_i}{\phi_r} = \frac{K_{PD}K_{VCDL}K_{LP}}{1 + K_{PD}K_{VCDL}K_{LP}} \equiv \frac{K}{1 + K}$$

$$\frac{\phi_e}{\phi_r} \equiv \frac{\phi_r + \phi_i - \phi_o}{\phi_r} = \frac{1}{1 + K} \approx \frac{1}{K}$$

1st Order DLL Transfer Function

- $H(s) = K_{LP}/s$,
- $BW = \text{loop gain}$



Loop BW

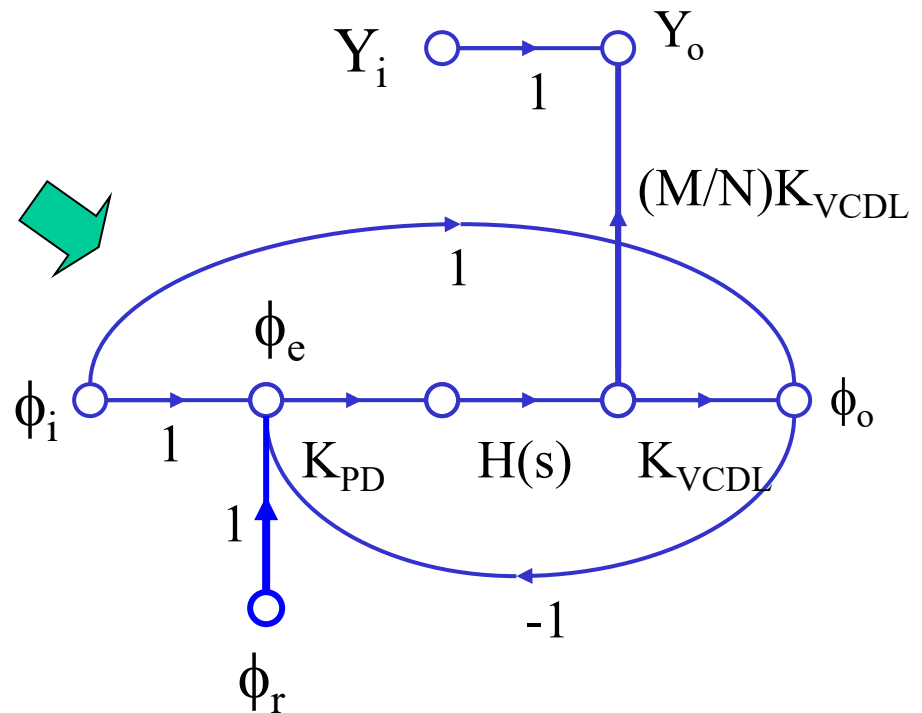
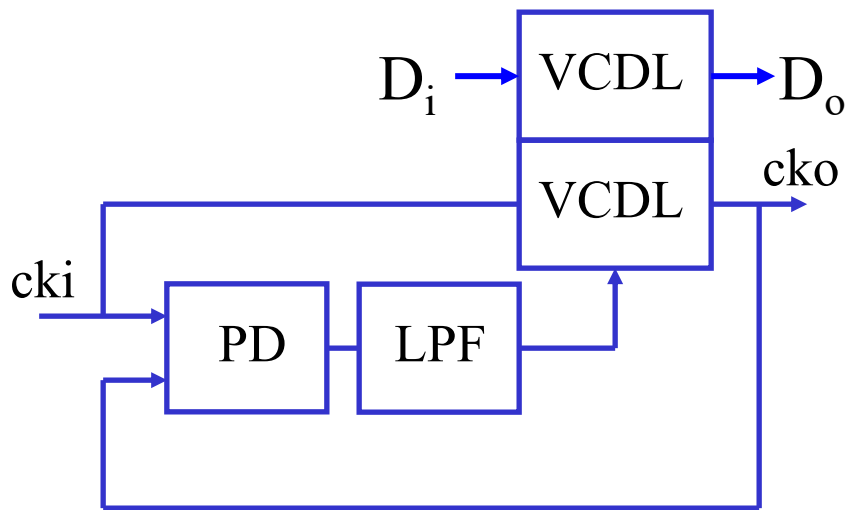
$$\frac{\phi_o - \phi_i}{\phi_r} = \frac{K_{PD}K_{VCDL}K_{LP}/s}{1 + K_{PD}K_{VCDL}K_{LP}/s} \equiv \frac{1}{1 + s/K}$$

$$\frac{\phi_e}{\phi_r} \equiv \frac{\phi_r + \phi_i - \phi_o}{\phi_r} = \frac{s/K}{1 + s/K}$$

Linear M-S DLL Circuit Model



- (M/N) : Slaver/Master delay line ratio

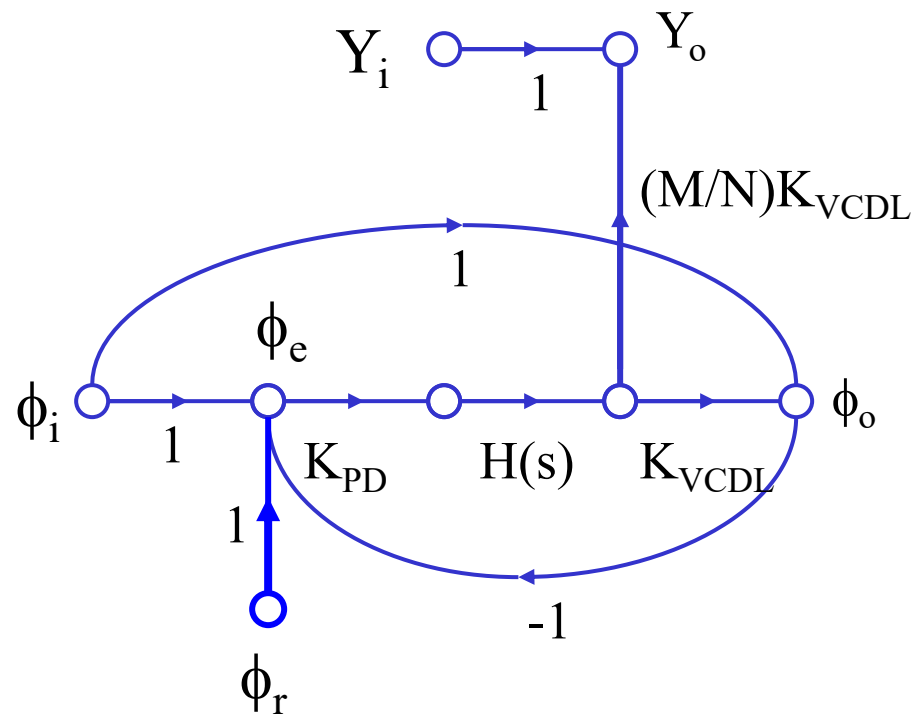


Linear M-S DLL Circuit Model



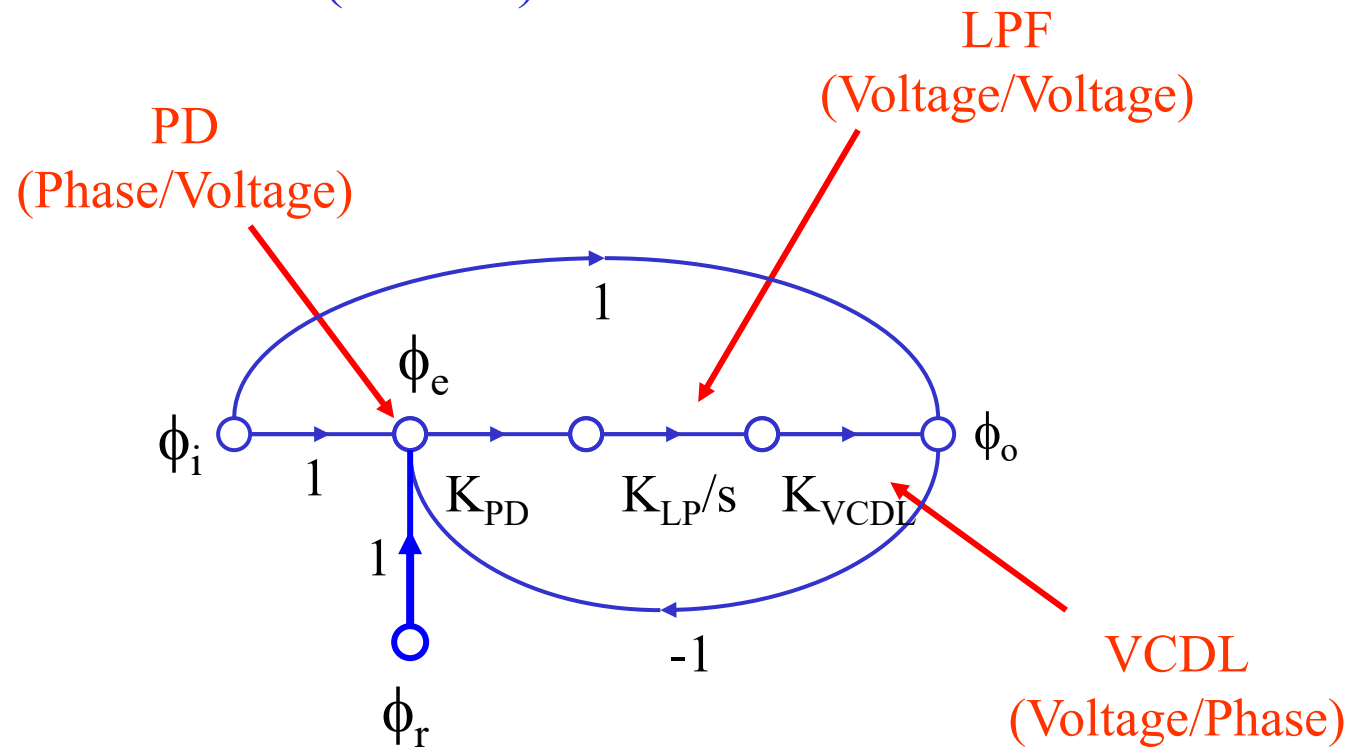
- (M/N): Slaver/Master delay line ratio

$$\frac{Y_o - Y_o}{\phi_r} = \frac{N}{M} \frac{K_{PD} K_{VCDL} H(s)}{1 + K_{PD} K_{VCDL} H(s)}$$



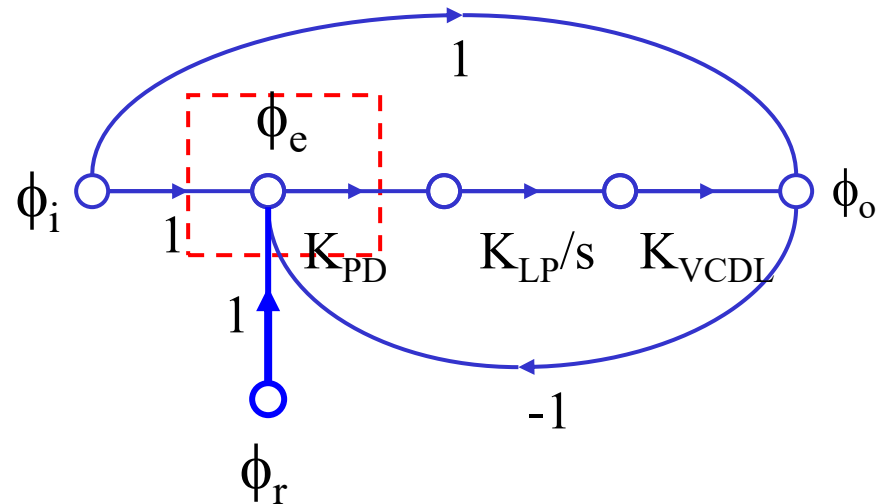
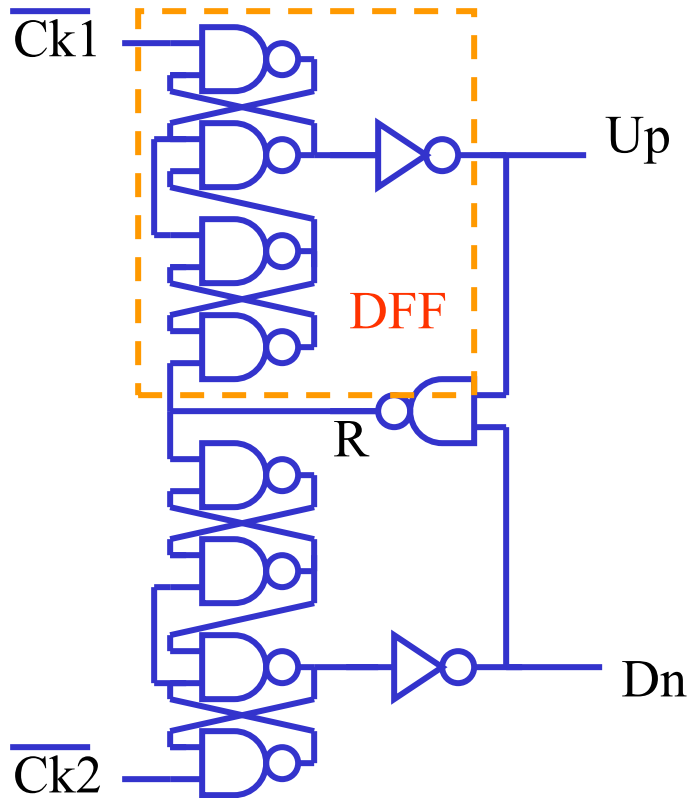
VLSI DLL Circuit Implementation

- Phase Detector(PD)
- Loop Filter (LPF)
- Voltage Control VCDL (VCDL)



VLSI Phase Detector Implementation

- Phase Detector Circuits



VLSI Phase Detector Implementation

- Phase Detector Circuit Implementation

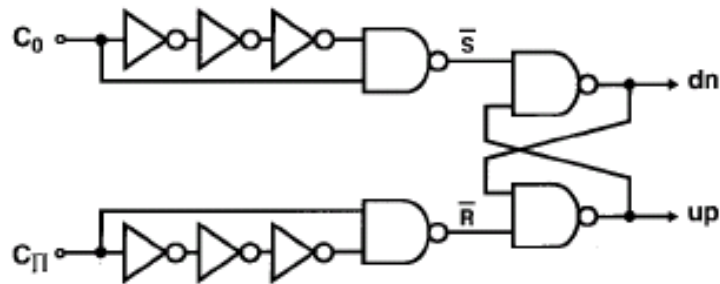
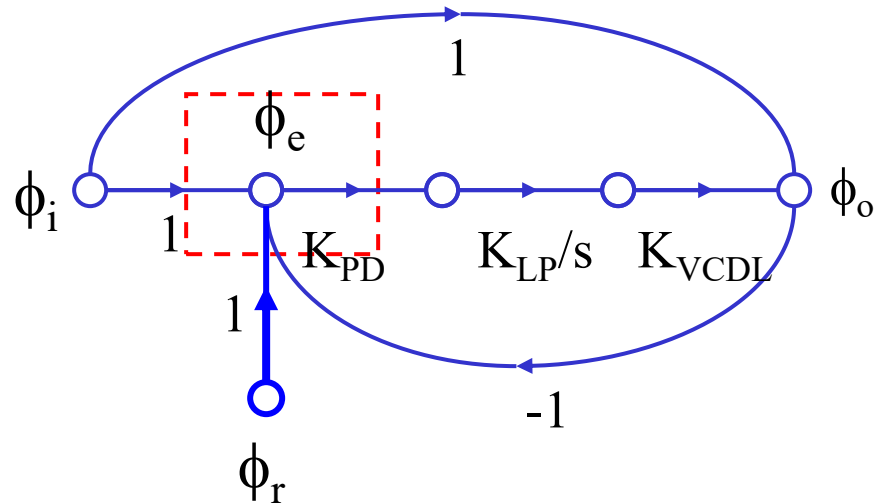
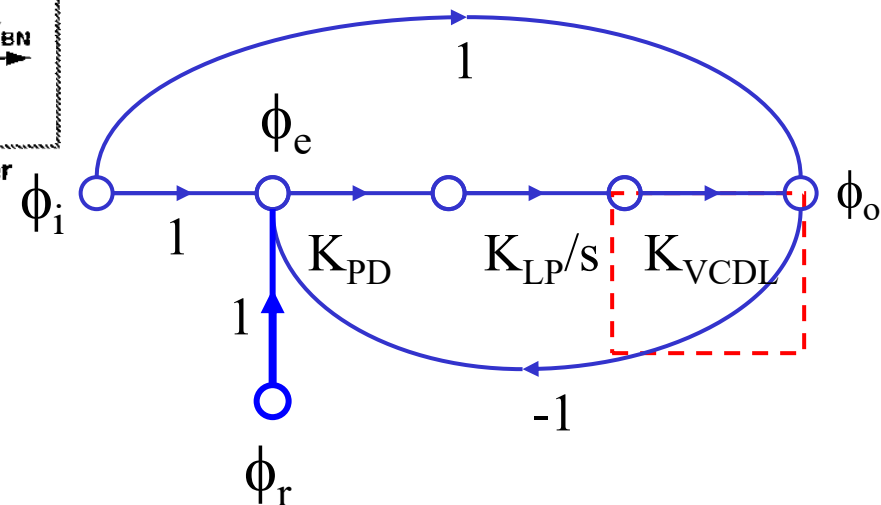
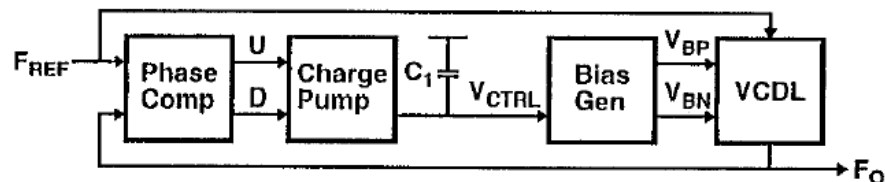
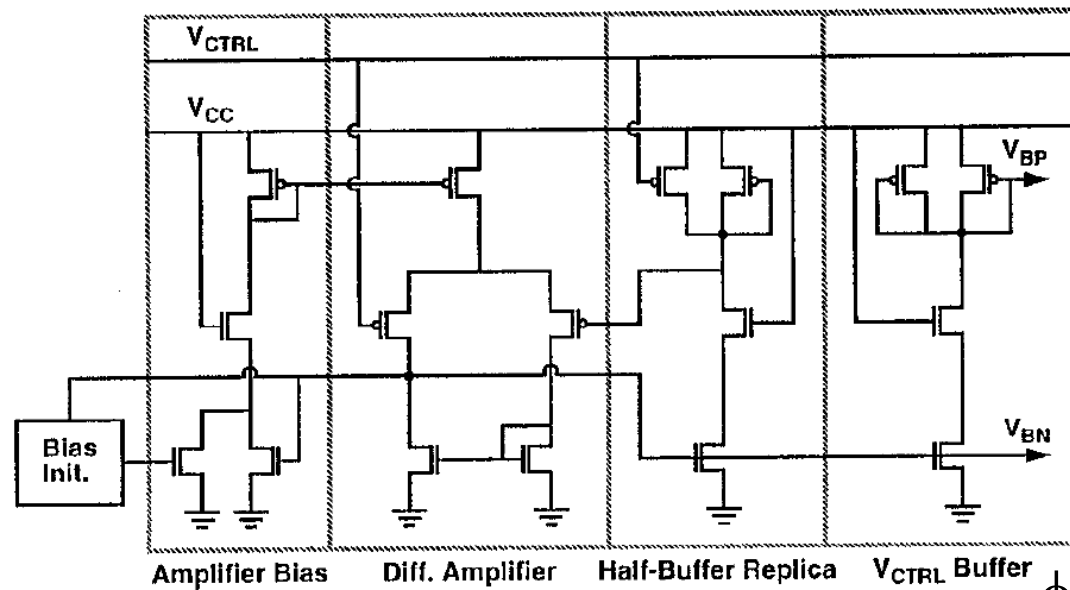


Fig. 7. Core loop phase detector.



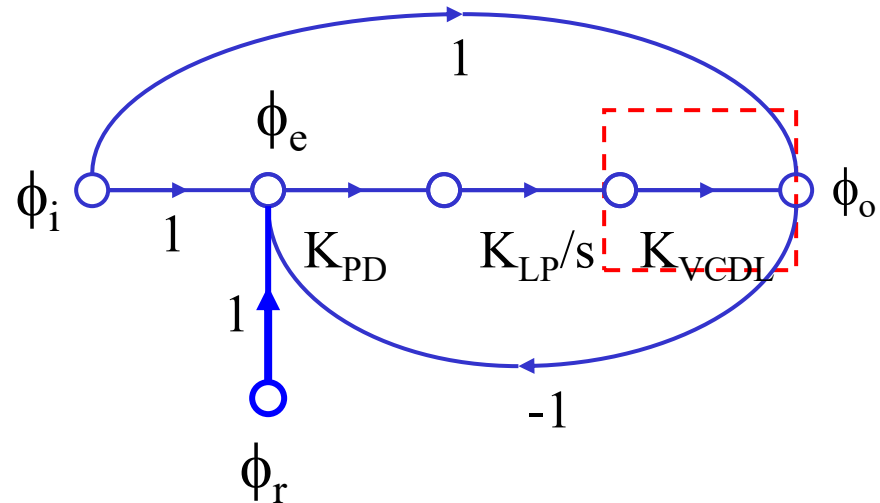
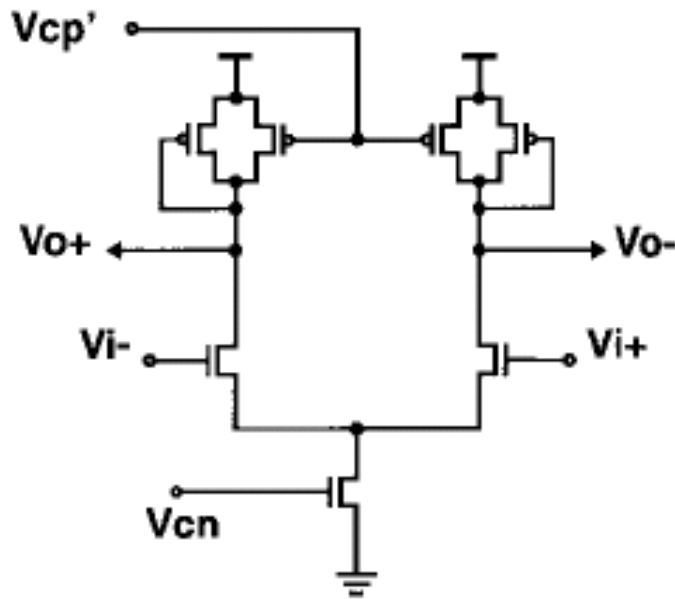
VLSI VCDL Implementation

- Symmetrical Load Delay Line Replica Biasing



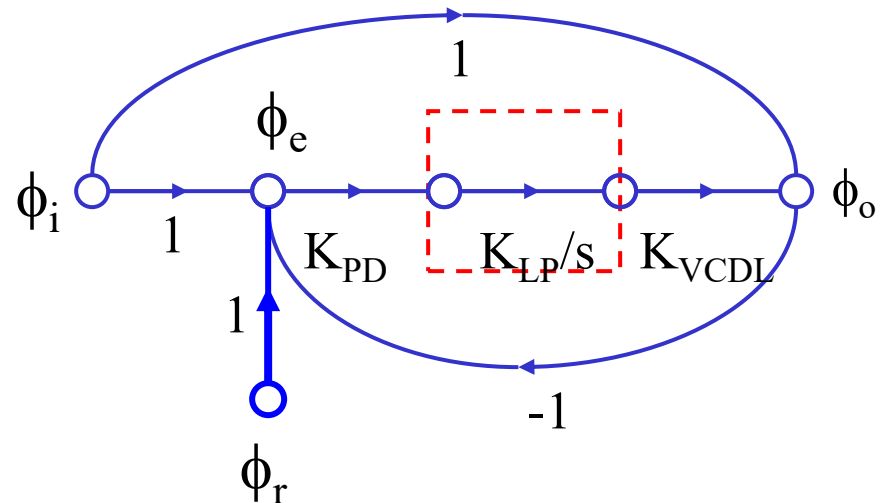
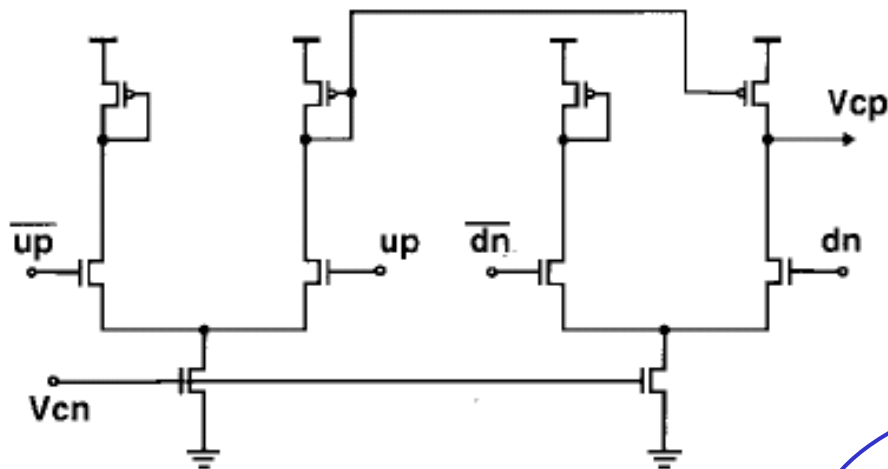
VLSI VCDL Implementation

- Symmetrical Load Delay Elements

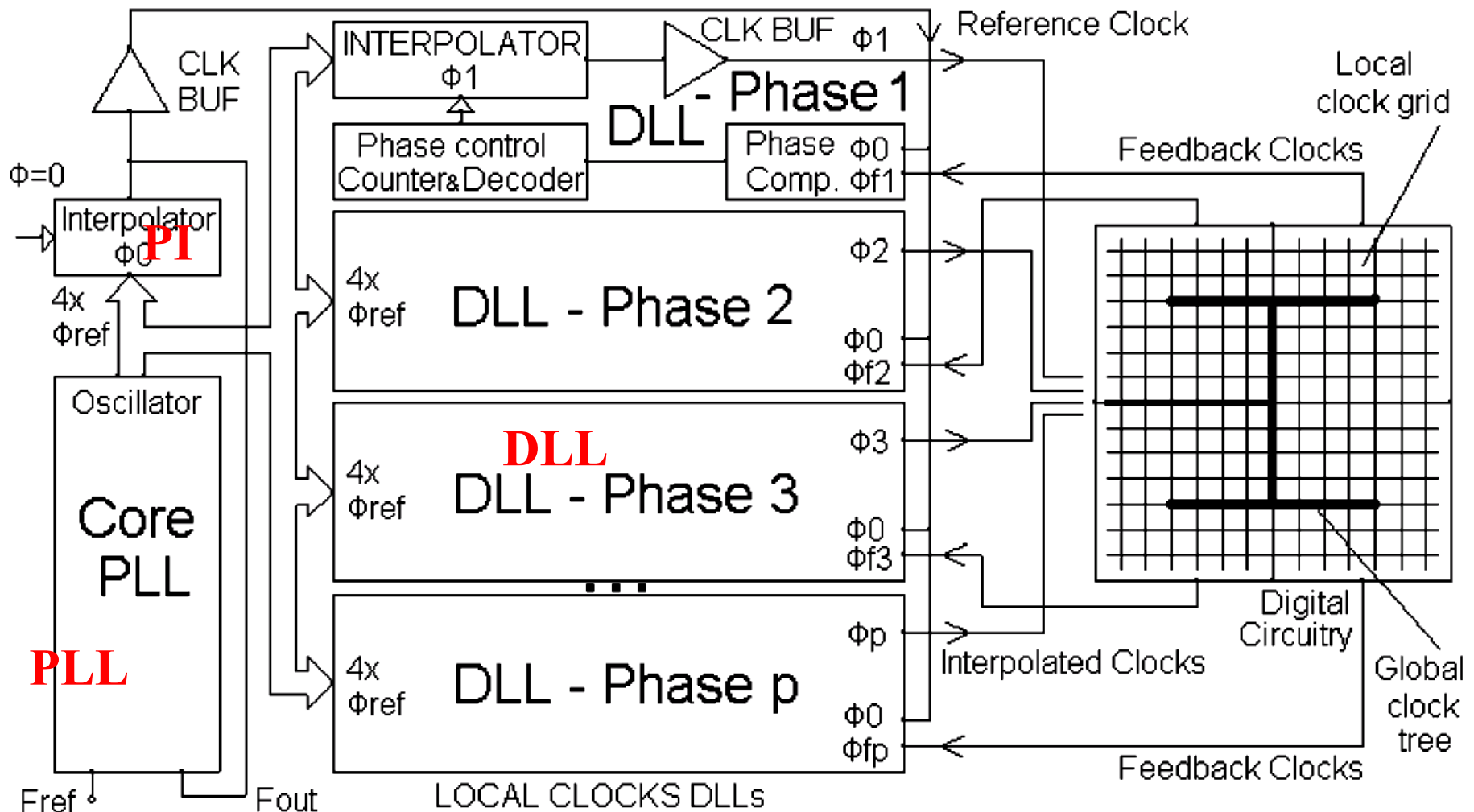


VLSI Loop Filter Implementation

- Charge Pump Circuit



DLL Application: CPU Clock Tree



Reference



- [Man96] John G. Maneatis, “ Low-Jitter Process-Independent DLL and PLL Based on Self-Biased Techniques,” IEEE Journal of Solid-State Circuits, Vol.31, Nov. 1996.
- [Kim02] C. Kim, et al, “Low-Power Small-Area +/-7.28ps Jitter 1Ghz DLL-Based Clock Generator,” ISSCC 2002.
- [Song0f] H.J. Song, “Adaptive DE-Skew Clock Generation”, US Patent #6,917,660, 200f.

Reading Assignment



- Please read DLL Chapter of the textbook